

# S. A. I. Shouborno

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## Research interests

Interpretable and physics-informed machine learning for sensor data; wearable health sensing, including cuffless blood pressure and sleep-based cognitive screening; industrial cyber-physical systems; language models grounded in sensor data; efficient on-device inference.

## Education

**Ph.D., Electrical and Computer Engineering**, University of Massachusetts Amherst Aug 2023 – May 2028 (expected)  
Advisor Bashima Islam. Began at Worcester Polytechnic Institute in Aug 2023; transferred with the lab in 2026.

**M.S., Electrical and Computer Engineering**, Worcester Polytechnic Institute Aug 2023 – Dec 2025  
GPA 4.00/4.00. Thesis: Interpretable and Efficient AI for Sensor-Rich Mobile and Embedded Systems.

**B.Sc., Electrical and Electronic Engineering**, Bangladesh University of Engineering and Technology Feb 2016 – Feb 2021  
Thesis: Audiovisual emotion recognition using multi-head attention based neural network, advised by Celia Shahnaz.

## Experience

**Research Assistant, University of Massachusetts Amherst, Amherst, MA** Sep 2026 – Present

- Building physics-informed neural networks that estimate blood pressure from graphene electronic tattoo bioimpedance, with the Kireev Lab.
- Extending the Bosch internship project with Bosch Research by reconstructing the program execution traces of industrial controllers from process sensors (IAAI-27, under review). The method detects code-path faults that sensor readings alone cannot reveal at 0.99 AUC and names the faulty function 69% of the time.
- Predicted future cognitive impairment from overnight sleep studies at 0.77 age-conditioned AUROC on held-out hospitals in the PhysioNet Challenge 2026 (CinC 2026, accepted).

**AI for Cyber-Physical Systems Intern, Robert Bosch LLC, Bosch Research, Pittsburgh, PA** May 2025 – Aug 2025

- Built a contrastive model aligning vision, sensor, program trace and text data from a manufacturing process for language-guided few-shot classification of machine state.

**Research and Teaching Assistant, Worcester Polytechnic Institute, Worcester, MA** Aug 2023 – Aug 2026

- Developed LLaSA, a sensor-aware LLM that answers questions about human activity from IMU data and outperforms GPT-3.5 and GPT-4o-mini, with 235,661 released training pairs (SensorCap and OpenSQA).
- Developed JOLT, an early-exit training method that ranked first of 11 methods across 7 datasets under compute budgets.
- Served as teaching assistant for the graduate course On-Device Deep Learning and three undergraduate ECE courses.

**Machine Learning Engineer, Hishab Technologies, Dhaka, Bangladesh** Feb 2021 – Jul 2023

- Built telephony conversational AI for Bengali speakers, covering speech recognition, text to speech, named entity recognition, language models and grapheme-to-phoneme conversion.
- Deployed services for banking and microfinance clients, including Bank Asia and Hitachi India.

## Publications

### Under review

S. A. I. Shouborno, Emily Ruppel, Sudershan Boovaraghavan, Bashima Islam. **Program Execution Traces Without Instrumentation: Debugging Industrial Controllers from Physical Process Sensors**. AAAI Conference on Innovative Applications of Artificial Intelligence (IAAI-27). [code]

### Journal articles

T. I. Mahmud, S. A. Imran, C. Shahnaz. **Res-SE-ConvNet: A Deep Neural Network for Hypoxemia Severity Prediction for Hospital In-Patients Using Photoplethysmograph Signal**. IEEE Journal of Translational Engineering in Health and Medicine, vol. 10, pp. 1–9, 2022. [code]

A. B. A. Qayyum, K. M. N. Hassan, A. Anika, M. F. Shadiq, M. M. Rahman, M. T. Islam, S. A. Imran, S. Hossain, M. A. Haque. **DOANet: A Deep Dilated Convolutional Neural Network Approach for Search and Rescue with Drone-Embedded Sound Source Localization**. EURASIP Journal on Audio, Speech, and Music Processing, vol. 2020, no. 1, art. 16, 2020. [code]

## Conference papers

- S. A. I. Shouborno, Mohammad Nur Hossain Khan, Bashima Islam. **JOLT: Jointly Optimized Layerwise Termination for Budgeted Early-Exit Inference**. IEEE International Conference on Tools with Artificial Intelligence (ICTAI). **Accepted, 2026**. [\[code\]](#)
- S. A. I. Shouborno, Bashima Islam. **Cross-Site Generalization of Interpretable Sleep Features for Predicting Future Cognitive Impairment**. Computing in Cardiology (CinC). **Accepted, 2026**. [\[code\]](#)
- S. A. Imran, M. T. Islam, C. Shahnaz, M. T. Islam, O. T. Imam, M. Haque. **COVID-19 mRNA Vaccine Degradation Prediction Using Regularized LSTM Model**. IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE), pp. 328–331, 2020. [\[code\]](#)
- O. T. Imam, M. Haque, C. Shahnaz, S. A. Imran, M. T. Islam, M. T. Islam. **Detection of Tuberculosis from Chest X-Ray Images Based on Modified Inception Deep Neural Network Model**. IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE), pp. 360–363, 2020.
- M. T. Islam, S. A. Imran, A. Arefeen, M. Hasan, C. Shahnaz. **Source and Camera Independent Ophthalmic Disease Recognition from Fundus Image Using Neural Network**. IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON), pp. 59–63, 2019.
- A. B. A. A. Qayyum, A. Anika, M. M. M. Miah, M. M. Rahman, K. M. N. Hasan, M. T. Islam, S. A. I. Shouborno, M. F. Shadiq, M. A. Haque. **Direction of Arrival Estimation through Noise Suppression: A Novel Approach Using GSC Beamforming and Room Acoustic Simulation**. IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON), pp. 104–108, 2019.
- S. A. I. Shouborno, T. I. Mahmud, N. Ishraq, R. Ali, T. H. Joy, S. A. Fattah, A. Bhattacharjee, C. Shahnaz. **Complete Automation of an E-commerce System with Internet of Things**. IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON), pp. 81–86, 2019.

## Workshop and short papers

- Tyler Bliss, Mahit Verma, Aila Iyer-Singh, Subrata Biswas, **Sheikh Asif Imran**, Bashima Islam. **Concord: An Agreement-Aware Multi-Adjudication Pipeline for LLM Evaluation**. Proceedings of the Fifth Workshop on Generation, Evaluation and Metrics (GEM), pp. 502–510, 2026.
- S. A. I. Shouborno, Mohammad Nur Hossain Khan, Subrata Biswas, Bashima Islam. **LLaSA: A Sensor-Aware LLM for Natural Language Reasoning of Human Activity from IMU Data**. Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion), pp. 893–899, 2025. [\[code and datasets\]](#)

## Patent

T. Al Muntasir, M. T. Islam, S. A. Imran. **System and Method for Optimizing a User Interaction Session within an Interactive Voice Response System**. U.S. Patent Application No. 18/378,118, 2025.

## Honors and awards

|                                                                                                             |            |
|-------------------------------------------------------------------------------------------------------------|------------|
| 2nd place (ICIP 2019) and 5th place (ICIP 2020), IEEE Video and Image Processing Cup                        | 2019, 2020 |
| 4th place, IEEE Signal Processing Cup at ICASSP 2020                                                        | 2020       |
| National champion and 22nd worldwide, IEEEExtreme 12.0 programming contest (also national champion in 2017) | 2018       |

## Service

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Chair, IEEE Signal Processing Society Student Chapter, BUET</b>                                                                                                                                                                        | 2019 – 2021 |
| <ul style="list-style-type: none"><li>Organized the 2019 IEEE SPS Winter School on Multimodal Signal Processing.</li><li>Grew it into the largest IEEE student branch chapter in Bangladesh, earning an IEEE SPS growth reward.</li></ul> |             |
| <b>Ambassador and Country Lead (2019), IEEEExtreme Programming Contest</b>                                                                                                                                                                | 2017 – 2019 |

## Skills

**Programming:** Python, PyTorch, TensorFlow, Keras, scikit-learn, MATLAB, C/C++, Git, LaTeX

**Machine learning:** Multimodal and contrastive learning, LLM fine-tuning, on-device inference, physics-informed models

**Signals:** Biomedical signals (PPG, bioimpedance, sleep PSG), IMU, audio and speech, image and video

**Systems:** SLURM clusters, MLOps, GCP and AWS, Kafka, Jenkins, microcontrollers, FPGA

## References

Available on request.